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Proposal Report for
‘HIGHWAY DESIGN PROJECT’

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Contents

INDIVIDUAL CONTRIBUTION

Name	Reg.Number	Contribution
W.M.V.O.L Thilakarathne	D/ENG/24/0074	Traffic Survey, Traffic Analysis
M.K.C.K Madawala	D/ENG/24/0077	Geometric Design Improvements, Engineering Drawing
C.D.H Liyanage	D/ENG/24/0099	Pavement Design
K.K.V.Y.A Vidanage	D/ENG/24/0104	Introduction, objectives, study area
M.F.H Ahamed	D/ENG/24/0112	LOS Analysis, Subgrade Strength Evaluation
M.M.A Ahamed	D/ENG/24/0121	Road Inventory Survey

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1. INTRODUCTION

The proposed Highway Design Project for the **Gangarama Road** aims to provide practical experience in the application of highway engineering principles while contributing to the improvement of local road infrastructure and connectivity within the **Boralesgamuwa** area. This project focuses on evaluating and redesigning existing road segment to enhance traffic flow, safety, and accessibility, thereby supporting the increasing transportation demands of the region. As a developing suburban area, Boralesgamuwa experiences a steady growth in traffic volume due to residential, commercial, and institutional activities, making it essential to upgrade existing road facilities.

The selected road segment plays an important role in facilitating the movement of people and goods within the area, acting as a key link between surrounding neighborhoods and nearby urban centers. However, the current condition and geometric characteristics of the road may not be adequate to handle future traffic demand efficiently. Therefore, this project is designed to assess existing road conditions and identify deficiencies related to capacity, safety, pavement conditions, and drainage.

To address these issues, the study includes detailed field investigations such as road inventory surveys, traffic data collection, and subgrade strength evaluation. These analyses will help determine the current performance of the road, including its Level of Service (LOS). Based on the findings, appropriate design improvements will be proposed, including geometric design enhancements, flexible pavement design, and drainage system development.

Overall, this project aims to develop a sustainable and efficient road design that accommodates future traffic growth over a 15-year design period. It also provides an opportunity to integrate theoretical knowledge with practical application, ensuring that the redesigned roadway meets modern engineering standards while improving safety, durability, and user comfort.

2. OBJECTIVES

- To evaluate the existing condition of the selected road segment (**Gangarama Road**) through field surveys and observations.
- To determine the current traffic characteristics, including traffic volume, composition, and peak hour conditions.
- To assess the present performance of the road by determining its Level of Service (LOS).
- To estimate future traffic demand for a design period of 15 years by considering traffic growth rates.
- To redesign the road segment to accommodate future traffic requirements efficiently and safely.
- To design a suitable flexible pavement structure based on traffic loading and subgrade strength using standard design methods.
- To evaluate and improve geometric features such as lane width, shoulder width, alignment, and sight distance.
- To design an effective drainage system, including culverts, to prevent water-related damage and ensure durability of the road.
- To apply standard highway engineering principles and practices in a practical, real-world situation.
- To develop engineering drawings and present results clearly with proper calculations, assumptions, and justifications.

3. SITE INVESTIGATION AND PRELIMINARY INVESTIGATION

3.1 Study Area

The study area selected for this project is a section of **Gangarama Road**, located near **Boralesgamuwa** town Colombo district. This road segment is a two-lane roadway with one lane in each direction and a total length of approximately 1 km to 1.5 km, as required by the project guidelines. The selected section includes at least one drainage crossing location, which is essential for drainage and culvert design analysis.

Figure 1: Project area

3.2 Road Inventory Survey

A field survey was conducted along the selected section of Gangarama Road to collect essential data regarding the existing physical, geometric, and operational characteristics of the roadway. Observations were made through direct measurement and visual inspection, supported by site photographs.

3.2.1 Carriageway Width

The measured carriageway width of the road is **7.4 m**. The road operates as a narrow two-way facility, and this width is just adequate for vehicle movement but may cause difficulty when two vehicles pass simultaneously.

3.2.2 Shoulder Width

The shoulder width was measured as **0.5 m**. The shoulder is mostly earthen and provides limited space for emergency stopping and pedestrian use. In several locations, the shoulder condition is uneven and partially eroded.

3.2.3 Number of Lanes

The road consists of a **two-lane, two-way configuration**, with one lane provided for each direction of traffic.

3.2.4 Horizontal Alignment

Horizontal alignment refers to the layout of the road in plan, consisting of straights and curves. In the study section, the road consists of **straight segments combined with gentle curves**. These curves influence vehicle speed and reduce overtaking opportunities at certain locations.

3.2.5 Vertical Alignment

Vertical alignment represents the profile of the road along its length. The observed road section shows a combination of **flat and slightly sloping sections**. These variations are not severe but can influence vehicle movement and drainage patterns.

3.2.6 Sight Distance Conditions

Sight distance is the visible length of roadway ahead available to a driver. It is an important factor for safe driving and includes stopping and overtaking distances.

In the study section:

- Sight distance is **clear in straight sections**
- Sight distance is **restricted at curved sections and built-up areas**

From the photos, visibility is reduced due to:

- roadside buildings
- parked vehicles
- vegetation
- narrow road width

This condition may affect safety stopping and overtaking.

3.2.7 Roadside Features

The roadside environment consists of:

- **houses and boundary walls**
- **shops and commercial buildings**
- **trees and vegetation**
- **utility poles and parked vehicles**

These features indicate a mixed residential and commercial area. Roadside activities and obstructions reduce lateral clearance and affect traffic flow and safety.

3.2.8 Existing Drainage System

The drainage system is **partially available** along the road.

From observations:

- Some sections have **functional side drains**
- Some areas are **blocked or poorly maintained**
- In certain locations, no proper drainage is visible

Blocked drainage and shoulder erosion indicate poor water flow, which can damage the pavement over time.

3.2.9 Pavement Condition

The pavement condition was assessed visually. The road surface shows signs of deterioration, including:

- **cracks**
- **potholes**
- **edge wear**

These defects affect riding quality and indicate the need for maintenance or rehabilitation.

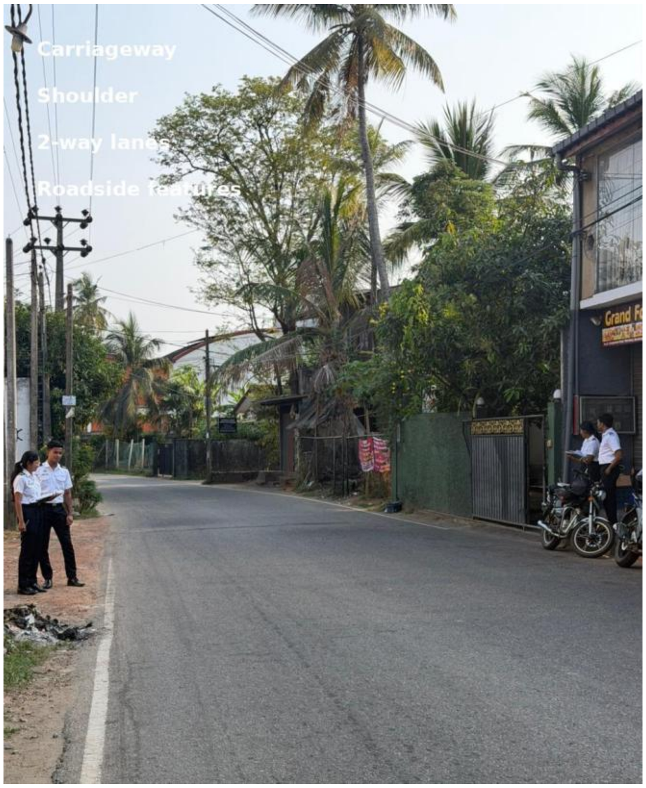


Figure 3:Pavement surface condition showing cracks and surface wear

Figure 2: Horizontal curve affecting sight distance and vehicle movement

Figure 5: Typical Road section showing carriageway, shoulder, two-way traffic lanes, and roadside features

Figure 4: Shoulder width

3.3 Traffic Survey

The traffic data presented below were collected on a weekday between 4.00pm to 6.00pm in Gangaramaya road. Total number of 6 vehicle types were observed during the survey.

Table 1: Boralasgamuwato Borupana

Vehicle Type	4.00-4.15	4.15-4.30	4.30-4.45	4.45-5.00	5.00-5.15	5.15-5.30	5.30-5.45	5.45-6.00
Bike	39	47	59	66	72	69	81	90
Car	17	21	35	38	41	37	40	52
Three Wheel	31	33	38	41	43	50	47	52
Suv + Van	07	15	23	34	36	40	33	44
6-wheel lorry	-	04	03	05	05	02	03	05
Bus	-	-	-	-	-	01	-	-
Total	94	120	158	184	197	199	204	244

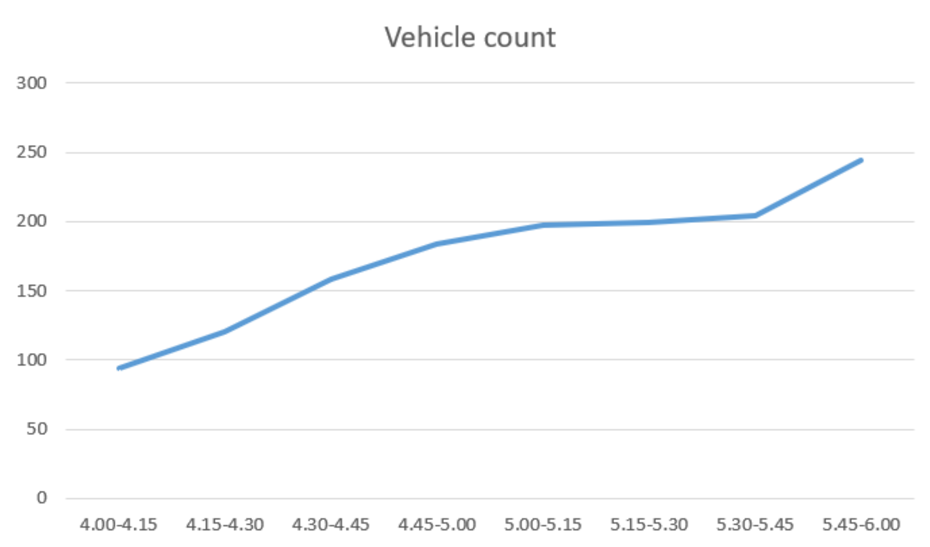


Figure 6: Vehicle Count

Table 2: Borupana to Boralasgamuwa

Vehicle Type	4.00-4.15	4.15-4.30	4.30-4.45	4.45-5.00	5.00-5.15	5.15-5.30	5.30-5.45	5.45-6.00
Bike	43	52	61	70	68	85	90	96
Car	26	31	30	35	37	46	42	56
Three Wheel	26	37	32	40	47	52	50	55
Suv + Van	09	11	21	35	26	33	26	34
6-wheel lorry	03	03	01	-	03	04	-	05
Bus	-	-	-	-	-	01	-	-
Total	107	134	145	181	181	221	208	246

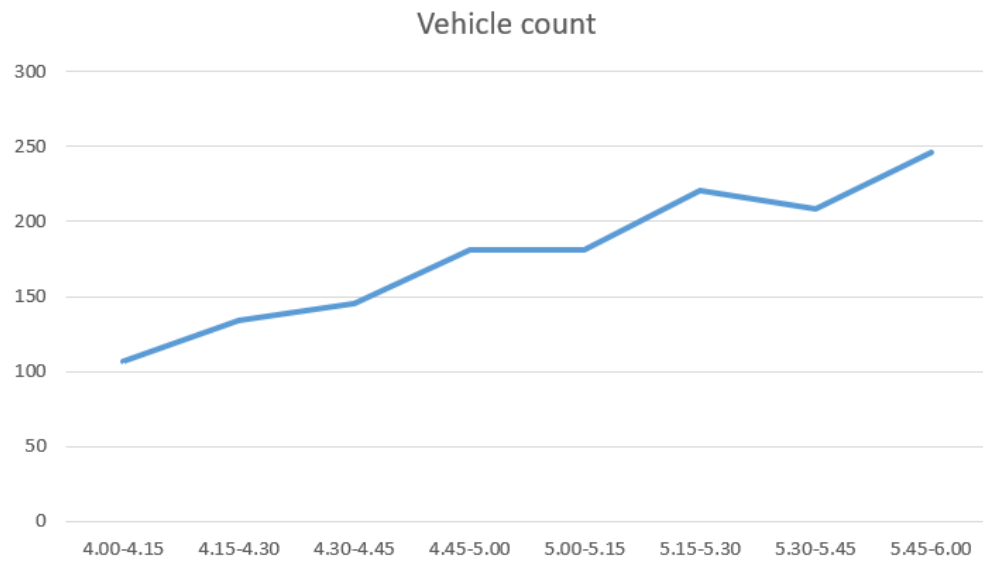


Figure 7: Vehicle Count

3.4 Traffic Analysis

Considering peak hour as 5.00pm – 6.00pm

Vehicle amount of

1st 15 minutes - 181

2nd 15 minutes - 221

3rd 15 minutes - 208

4th 15 minutes – 246

Peak hour volume = 856

Peak hour factor (PHF) $= \frac{856}{246 \times 4} = 0.87$

Heavy Vehicle Percentage

Heavy vehicles include buses, two-axle trucks (6 tires), and vehicle-carriage combinations.

To Borupana road

- Heavy vehicles = 1(buses) + 28(2-axle) = 29
- Total vehicles = 1400

$$HV\% = \frac{29}{1400} \times 100 = 2.1\%$$

To Boralasgamuwa

- Heavy vehicles = 1 (buses) + 19 (2-axle) = **20**
- Total vehicles = 1423

$$HV\% = \frac{20}{1423} \times 100 = 1.4\%$$

Overall Heavy Vehicle Percentage

- Total heavy vehicles = 29 + 20 = **49**
- Total traffic volume = 1400 + 1423 = **2823**

$$HV\% = \frac{49}{2823} \times 100 = 1.7\%$$

Traffic Growth Projection

A traffic growth rate of **4% per annum** is assumed as per assignment guidelines, with a design life of **15 years**.

The future traffic growth can be estimated using the compound growth formula:

$$T_f = T_0(1 + r)^n$$

T_f − Future traffic

T_0 − Present traffic

r – annual growth rate as decimal

n – number of years

Therefore, for a growth rate of 4% with a design life of 15 years, we can use the following equation

$$T_f = T_0(1 + 0.04)^{15}$$

$$T_f = T_0(1.004)^{15}$$

3.5 Level of Service (LOS) Analysis

Capacity analysis estimates how much traffic a road can carry under specific conditions. Road capacity is the maximum number of vehicles that can pass a point or road section within an hour under existing traffic, roadway, and control conditions. The Transportation Research Board's Highway Capacity Manual (USA) is commonly used as a guideline for highway design and evaluation.

The concept of Level of Service (LOS) is the different operational qualities of the traffic flow. Level of Service is a qualitative measure of speed and travel time, freedom to maneuvers, traffic interruption, driving comfort, convenience, safety and operating cost.

Six Levels of Service are categorized from A to E, and it covers entire range of traffic conditions which may occur

Level of Service A - Free flow at high speed and low volumes (Drivers can hold their desired speeds without delays)

Level of Service B - Reasonably free flow (Stable flow and drivers have reasonable freedom to select their speed)

Level of Service C - Stable flow (Most drivers are restricted in their freedom to select their own speed, change lanes, overtake etc.)

Level of Service D – Borderline Unstable (Comfort and convenience are low but may be tolerated for short periods)

Level of Service E – Extremely Unstable (This LOS is obtained with traffic volumes with near or at capacity)

3.5.1 Required values for analysis

Peak Hour Volume

Consider Peak Hour as 5pm – 6pm

Peak Hour volume = 856

Peak hour factor = 856/(246x4) = 0.87

v = V/PHF = 856/0.87 = 984

Passenger Car Unit (PCU)

Passenger Car Unit (PCU) is a vehicle unit or car unit used to measure the rate of traffic flow on highway. In other words, PCU is a measure of number of vehicles moving on a highway at a given point of time. In some instances, PCU is referred to as Passenger Car Equivalent (PCE).

Traffic flow is a measure of flow of vehicles on a road from one point to another point at a given point of time. The flow of traffic on highway consists of different classes of vehicles that are classed as follows:

Passenger Car, Small Bus, Large Bus, Light Truck, Medium Truck, Heavy Truck, Motorcycle, Bicycle, Animal Drawn Carts

Table 3: Equivalent Passenger Car Unit (PCU) Factors For two-lane, two-way roads

Vehicle Type	PCU Factor (Flat Terrain)
Passenger Car	1.0
Small Bus	2.0
Large Bus	2.2
Light Truck	2.0
Medium Truck	2.2
Heavy Truck	2.2
Motorcycle	0.5
Bicycle	1.0
Animal Drawn Carts	4.0

(Source: RDA, Traffic and Planning Division)

Table 4: PCU values for each vehicle type

Vehicle Type	Count (2hrs)	PCU Factor	PCU
Passenger Car	638	1.0	638.0
Motorcycle	1088	0.5	544.0
Three-Wheel	675	0.5	337.5
Other 2 Axel, 4 tires	427	1.0	427.0
2 Axel 6 Tyre	47	2.0	94.0
Bus	2	2.2	4.4

Total PCU in Peak Hour = (638.0+544.0+337.5+427.0+94.0+4.4)/2 = 1022.45 PCU/hr

Service Flow Rate

Service flow rate is the maximum rate of flow which can be accommodated by various facilities at each level of service A to E. It is the maximum hourly rate under prevailing conditions while maintaining the designed level of service.

To calculate the service flow rate, following equations are applied.

$$\boldsymbol{SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}}$$

$$\boldsymbol{f_{HV} = \frac{1}{[1 + P_T(E_T - 1) + P_B(E_B - 1)}}$$

$$\boldsymbol{f_g = \frac{1}{[1 + (P_p + I_p)]}}$$

$$\boldsymbol{I_p = 0.02(E - E_0)}$$

$$\boldsymbol{E_{HV} = 1 + (0.25 + P_{\frac{T}{HV}})(E - 1)}$$

SFi = Total service flow rate in both directions for prevailing roadway and Traffic conditions, for level of service i , in vph

$(v/c)_i$= ratio of flow rate to ideal capacity for LOS i or speed i obtained from Table 2.7

f_d= adjustment factor for directional distribution of traffic obtained from Table 2.8

f_w= adjustment factor for narrow lanes and restricted shoulder widths obtained from Table 2.5

f_g= adjustment factor for effects of grades on passenger cars

f_{HV}= adjustment factor for the presence of heavy vehicles in the traffic stream

P_P= proportion of passenger cars in the upgrade traffic stream, expressed as a decimal

I_P= impedance factor for passenger cars

P_{HV}= total proportion of heavy vehicles in the upgraded traffic stream

E_{HV}= Passenger Car Equivalent for specific mix on heavy vehicles present in the upgrade traffic stream

E = base passenger car equivalent for specific mix of heavy vehicles presents in the upgrade traffic stream (Table 2.9)

E_0= base passenger car equivalent for zero percent grade and a given speed selected from Table 2.19

$P_{T/HV}$= proportion of trucks among heavy vehicles

Volume-to-Capacity Ratio (v/c)

The volume-to-capacity ratio (v/c) represents the relationship between the traffic demand (volume) and the maximum capacity of a road. It indicates how efficiently a road is operating, with higher values showing more congestion. The (v/c)values are obtained from standard tables in the Highway Capacity Manual (, where each Level of Service (LOS) corresponds to a specific (v/c)value. The appropriate value is selected based on the desired or observed LOS and used in service flow calculations.

LOS	Percent time delay	Level Terrain							Rolling Terrain							Mountainous Terrain						
		Average[1] speed	% No passing zones						Average speed	% No passing zones						Average speed	% No passing zones					
			0	20	40	60	80	100		0	20	40	60	80	100		0	20	40	60	80	100
A	≤ 30	≥ 94	0.15	0.12	0.09	0.07	0.05	0.04	≥ 92	0.15	0.10	0.07	0.05	0.04	0.03	≥ 90	0.14	0.09	0.07	0.04	0.02	0.01
B	≤ 45	≥ 89	0.27	0.24	0.21	0.19	0.17	0.16	≥ 87	0.26	0.23	0.19	0.17	0.15	0.13	≥ 87	0.25	0.20	0.16	0.13	0.12	0.10
C	≤ 60	≥ 84	0.43	0.39	0.36	0.34	0.33	0.32	≥ 82	0.42	0.39	0.35	0.32	0.30	0.28	≥ 79	0.39	0.33	0.28	0.23	0.20	0.16
D	≤ 75	≥ 81	0.64	0.62	0.60	0.59	0.58	0.57	≥ 79	0.62	0.57	0.52	0.48	0.46	0.43	≥ 73	0.58	0.50	0.45	0.40	0.37	0.33
E	>75	≥ 73	1.00	1.00	1.00	1.00	1.00	1.00	≥ 65	0.97	0.94	0.92	0.91	0.9	0.90	≥ 57	0.91	0.87	0.84	0.82	0.80	0.78
F	100	< 45	-	-	-	-	-	-	< 40	-	-	-	-	-	-	< 35	-	-	-	-	-	-

Figure 8: Level of Service Criteria for General Two-Lane Highway Segments V/C Ratio (ratio of Flow Rate to an Ideal Capacity of 2800pcph in both Directions)

(*Source: HCM, Transport Research board, National Research council, Washington DC, USA)*

The volume-to-capacity ratio (v/c) represents the relationship between the traffic demand (volume) and the maximum capacity of a road. It indicates how efficiently a road is operating, with higher values showing more congestion. The (v/c)values are obtained from above table.

Table 5: v/c ratio

LOS	v/c
A	0.15
B	0.27
C	0.43
D	0.64
E	1.00

Adjustment factor for directional distribution of traffic (f_d)

the Adjustment factor for directional distribution factor (f_d) accounts for the proportion of traffic flow in each direction of a two-lane road. It reflects how traffic demand is split between the two directions during peak periods. In this analysis, a **50/50 distribution** is calculated, indicating equal traffic flow in both directions.

Calculation of Directional Distribution,

Directional Distribution = Peak Direction volume / Total Volume in Both Direction

= 1423/ (1423+1400)

= 0.5

So, the Directional Distribution is 50/50

Directional Distribution	100/0	90/10	80/20	70/30	60/40	50/50
Adjustment Factor, f_d	0.71	0.75	0.83	0.89	0.94	1.00

Figure 9: Adjustment Factors for Directional Distribution (fd) for Two Lane Highways

(Source: HCM, Transport Research board, National Research council, Washington DC, USA)

Therefore, the value of f_d is taken as **1.0** according to the above table. representing balanced traffic conditions.

Adjustment factor for narrow lanes and restricted shoulder widths (f_w)

The adjustment factor f_w accounts for the effect of lane width and shoulder width on road capacity. Wider lanes and shoulders provide better maneuvering space, improving traffic flow and safety. In Our case, a lane width of 3.7 m and a shoulder width of 1.8 m are measured, which represent standard design conditions.

Usable Shoulder Width (m)	3.7 m Lanes		3.4 m Lanes		3.0 m Lanes		2.7 m Lanes	
	LOS A - D	LOS E	LOS A - D	LOS E	LOS A - D	LOS E	LOS A - D	LOS E
>=1.8	1.00	1.00	0.93	0.94	0.84	0.87	0.70	0.76
1.2	0.92	0.97	0.85	0.92	0.77	0.85	0.65	0.74
0.6	0.81	0.93	0.75	0.88	0.68	0.81	0.57	0.70
0	0.70	0.88	0.65	0.82	0.58	0.75	0.49	0.66

Figure 10: Adjustment Factors for Combined Effect of Narrow Lanes and Restricted Shoulder Width (fw)

(Source: HCM, Transport Research board, National Research council, Washington DC, USA)

Therefore, the value of f_w is taken as 1.0 for LOS A – D and LOS E, indicating no reduction in capacity due to adequate geometric dimensions.

Adjustment factor for the effects of the grade (f_g)

The grade adjustment factor f_g was taken as 1.0, as the segment is considered as level terrain with negligible gradient. Under such conditions, there is no significant impact of slope on traffic flow, and hence no adjustment to capacity is required.

Adjustment factor for the presence of heavy vehicles in the traffic stream (f_{HV})

This factor f_{HV} accounts for the impact of heavy vehicles such as buses and trucks on traffic flow. Since heavy vehicles move slower and require more space than passenger cars, they reduce the overall road capacity. The value of f_{HV} is calculated using the proportion of heavy vehicles and their passenger car equivalent values. A higher percentage of heavy vehicles results in a lower f_{HV}, indicating reduced traffic performance.

Calculation Approach for f_{HV}

In this section, the calculation is carried out in a clear, step-by-step manner to ensure better explanation and ease of verification.

Step 1: Calculate the Percentage of Buses and Trucks (P_B & P_T)

Total vehicles count in peak volume Direction – 1423
Total Trucks in Both Direction – 19
Total Buses in Both Direction – 01

$$Percentage\ of\ Buses = \frac{1}{1423} \times 100 = 0.07\ \%$$

$$Percentage\ of\ Trucks = \frac{19}{1423} \times 100 = 1.34\ \%$$

Step 2: Determine the E_T & E_B values

Vehicle Type	Level of Service	Type of Terrain		
		Level	Rolling	Mountainous
Trucks, E_T	A	2.0	4.0	7.0
	B and C	2.2	5.0	10.0
	D and E	2.0	5.0	12.0
Buses, E_B	A	1.8	3.0	5.7
	B and C	2.0	3.4	6.0
	D and E	1.6	2.9	6.5

Figure 11: Average Passenger Car Equivalent for Trucks and Busses on Two Lane Highways over different Terrain Segments

(Source: HCM, Transport Research board, National Research council, Washington DC, USA)

According to the above standard table E_T & E_B values are determined

Table 6: Determined Et and Eb values for LOS

LOS	Level Terrain	
	Trucks, E_T	Buses, E_B
A	2.0	1.8
B	2.2	2.0
C	2.2	2.0
D	2.0	1.6
E	2.0	1.6

Step 2: Find $\boldsymbol{f_{HV}}$

For LOS A,

$$f_{HV} = \frac{1}{[1 + P_T(E_T - 1) + P_B(E_B - 1)}$$

$$\mathrm{f_{HV}} = \frac{1}{[1 + 0.013(2 - 1) + 0.0007(1.8 - 1)}$$

$$\mathrm{f_{HV}} = 0.987$$

For LOS B & C,

$$f_{HV} = \frac{1}{[1 + P_T(E_T - 1) + P_B(E_B - 1)}$$

$$\mathrm{f_{HV}} = \frac{1}{[1 + 0.013(2.2 - 1) + 0.0007(2 - 1)}$$

$$\mathrm{f_{HV}} = 0.984$$

For LOS D & E,

$$f_{HV} = \frac{1}{[1 + P_T(E_T - 1) + P_B(E_B - 1)}$$

$$\mathrm{f_{HV}} = \frac{1}{[1 + 0.013(2 - 1) + 0.0007(1.6 - 1)}$$

$$\mathrm{f_{HV}} = 0.986$$

3.5.2 Calculation of Service Flow Rate

For LOS A,

$$SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}$$

$$\mathrm{SF_i} = 2800 \times 0.15 \times 1 \times 1 \times 1 \times 0.987$$

$$\mathrm{SF_i} = 414.54$$

For LOS B,

$$SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}$$

$$\mathrm{SF_i} = 2800 \times 0.27 \times 1 \times 1 \times 1 \times 0.984$$

$$\mathrm{SF_i} = 743.90$$

For LOS C,

$$SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}$$

$$\mathrm{SF_i} = 2800 \times 0.43 \times 1 \times 1 \times 1 \times 0.984$$

$$\mathrm{SF_i} = 1184.74$$

For LOS D,

$$SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}$$

$$\mathrm{SF_i} = 2800 \times 0.64 \times 1 \times 1 \times 1 \times 0.986$$

$$\mathrm{SF_i} = 1766.91$$

For LOS E,

$$SF_i = 2800 \times (v/c)_i \times f_d \times f_w \times f_g \times f_{HV}$$

$$\mathrm{SF_i} = 2800 \times 1 \times 1 \times 1 \times 1 \times 0.986$$

$$\mathrm{SF_i} = 2760.80$$

Table 7: Summary Table

LOS	v/c	f_d	$\mathrm{f_w}$	$\mathrm{f_g}$	$\mathrm{f_{HV}}$	$\mathrm{SF_i}$
A	0.15	1	1	1	0.987	414.54
B	0.27	1	1	1	0.984	743.90
C	0.43	1	1	1	0.984	1184.74
D	0.64	1	1	1	0.986	1766.91
E	1.00	1	1	1	0.986	2760.81

3.5.3 Analysis

Determine Capacity of Two-Lane Highways

A two-lane highway is a two-lane road having one lane for use by traffic in each direction. Passing slower vehicles require the use of opposing lanes when sight distances and gaps in opposing traffic stream warrant.

Three parameters are used to describe service quality for two lane highways.

- Average travel speed - Reflects the mobility function of two-lane highways
- Percent time delay - Reflects both mobility and access functions
- Capacity utilization - Reflects the access function

Ideal Conditions for Two Lane Highways

- Design speed is greater than or equal to 96kmph
- Lane width is greater than or equal to 3.7m
- Clear shoulder width is greater than or equal to 1.8m
- No ‘No passing zones’ in the highway

- Only passenger cars in the traffic stream
- A 50/50 directional distribution of traffic.
- No impediment to through traffic dye to traffic control or turning vehicles.
- Level terrain.

The capacity of a two-lane highway under ideal conditions is 2800 pcph total in both directions.

Current LOS Identification

Observed v/c ratio – 1022.45/2800 = 0.37

So, the v/c is 0.27 – 0.43 Range

Compared with v and SF_i , the v value (984) is approximately match with SF_i c value (1184.74)

So that, currently operates LOS as C.

Adequacy of the Road the Existing Road:

The two-lane highway operates at a peak hour v/c ratio of 0.37, which falls within the SF_i c value, resulting in a **Level of Service (LOS) C**. This indicates stable traffic flow with noticeable but acceptable delays under level terrain conditions. The existing road is adequate for current traffic volumes.

Percent Grade	Average upgrade speed (kmph)	Percent 'No Passing Zones'					
		0	20	40	60	80	100
3	89	0.27	0.23	0.19	0.17	0.14	0.12
	85	0.42	0.38	0.33	0.31	0.29	0.27
	81	0.64	0.59	0.55	0.52	0.49	0.47
	73	1.00	0.95	0.91	0.88	0.86	0.84
	69	1.00	0.98	0.97	0.96	0.95	0.94
	65	1.00	1.00	1.00	1.00	1.00	1.00
4	89	0.25	0.21	0.18	0.16	0.13	0.11
	85	0.40	0.36	0.31	0.29	0.27	0.25
	81	0.61	0.56	0.52	0.49	0.47	0.45
	73	0.97	0.92	0.88	0.85	0.83	0.81
	69	0.99	0.96	0.95	0.94	0.93	0.92
	65	1.00	1.00	1.00	1.00	1.00	1.00
5	89	0.21	0.17	0.14	0.12	0.10	0.08
	85	0.36	0.31	0.27	0.24	0.22	0.20
	81	0.57	0.49	0.45	0.41	0.39	0.37
	73	0.93	0.84	0.79	0.75	0.72	0.70
	69	0.97	0.90	0.87	0.85	0.83	0.82
	65	0.98	0.96	0.95	0.94	0.93	0.92
	57	1.00	1.00	1.00	1.00	1.00	1.00
6	89	0.12	0.10	0.08	0.06	0.05	0.04
	85	0.27	0.22	0.18	0.16	0.14	0.13
	81	0.48	0.40	0.35	0.31	0.28	0.26
	73	0.49	0.76	0.68	0.63	0.59	0.55
	69	0.93	0.84	0.78	0.74	0.70	0.67
	65	0.97	0.91	0.87	0.83	0.81	0.78
	57	1.00	0.96	0.95	0.93	0.91	0.90
	49	1.00	0.99	0.99	0.98	0.98	0.98
7	89	0.00	0.00	0.00	0.00	0.00	0.00
	85	0.13	0.10	0.08	0.07	0.05	0.04
	81	0.34	0.27	0.22	0.18	0.15	0.12
	73	0.77	0.65	0.55	0.46	0.40	0.35
	69	0.86	0.75	0.67	0.60	0.54	0.48
	65	0.93	0.82	0.75	0.69	0.64	0.59
	57	1.00	0.91	0.87	0.82	0.79	0.76
	49	1.00	0.95	0.92	0.90	0.88	0.86

Figure 12: Values of V/C Ratios Vs Speed, Percent Grade and Percent No Passing Zones for Specific Grades

(Source: HCM, Transport Research board, National Research council, Washington DC, USA)

From the above table, for a 3% grade and an average upgrade speed of 85 km/h, the corresponding percent no-passing zone adjustment factor is 0.42. This value indicates that under these grade and speed conditions, passing opportunities are moderately restricted, which reduces the effective capacity of the two-lane highway. The factor is applied to adjust the two-way flow rate in level of service calculations.

Improvement current LOS

However, especially given that lane width is already 3.7 m, exceeding typical geometric requirements for **LOS B**. To achieve one LOS level above (LOS B) and accommodate future growth, improvements should focus on providing paved above 2.0m shoulders and enhancing passing sight distance to 500 m, as lane width is already sufficient.

- Add shoulders - Above 2.0 m
- Improve alignment - Reduce sharp curves
- Remove roadside obstacles
- Improve sight distance as 500m

3.6 Subgrade Strength Evaluation

3.6.1 Subgrade Strength

The subgrade soil strength is measured by its CBR value. Flexibility is available to the designer to carry out the CBR test at the dry density and moisture content of his choice. Usually, the dry density is selected to simulate the conditions corresponding to the minimum degree of compaction and the worst moisture content likely in the field. The dry density selected depends on the compaction specifications adhered to, while the moisture content selected depends on climate and drainage controls adopted.

For low to medium trafficked roads or highly trafficked roads the subgrade is generally expected to be compacted to 100% of maximum dry density under ASTM (10lb rammer) or BS (Heavy 4.5kg rammer) modified conditions of compaction respectively. Therefore, to simulate these conditions it is suggested that the CBR test be carried out on a sample compacted to these densities.

3.6.2 CBR Test Result (Laboratory)

The California Bearing Ratio (CBR) test is a standard laboratory method used to evaluate the strength and bearing capacity of subgrade soil for pavement design. In this test, a soil sample is compacted at optimum moisture content, soaked for four days to simulate the worst-case saturated condition, and then penetrated by a standard plunger at a rate of 1.27 mm/min. The resistance measured at 2.5 mm and 5.0 mm penetration is compared to that of a standard crushed stone to obtain the CBR value (expressed as a percentage). The CBR test provides a conservative estimate of subgrade strength under critical moisture conditions and is widely used in pavement design methodology as AASHTO.

Table 8: Laboratory CBR Test Result

Penetration(mm)	CBR%
0.5	1.74
1.0	1.99
1.5	2.18
2.0	2.32
2.5	2.62
3.0	2.7
4.0	2.82
5.0	3.14
7.5	3.46
10	3.69
12.5	4.19

These are relatively low CBR values (2–4%), indicating a **weak subgrade**.

3.6.3 DCP Test Result (In-situ)

The Dynamic Cone Penetrometer (DCP) test is a rapid, in-situ field test used to assess subgrade strength and stiffness. The apparatus consists of a steel rod with a conical tip that is driven into the soil by dropping a fixed weight (typically 8 kg) from a constant height (typically 575 mm). The penetration depth per blow (mm/blow) is recorded, and this DCP index is correlated to CBR using empirical relationships such as the TRL formula: $Log_{10}(CBR) = 2.48 - 1.057 \times Log_{10}(DCPI)$. The DCP test provides a continuous strength profile with depth, allows for numerous tests across a site, and gives an estimate of in-situ (dry) subgrade strength for comparison with laboratory results.

Location – 01

DCP (mm/blow)	CBR%
4	72.71
9	25.89
5	54.73
11	20.06
8	30.09
Location 1	

Table 9: In-situ DCP Result at Location 1

Location – 02

DCP (mm/blow)	CBR%
4	64.66
10	22.54
4	64.66
8	29.13
12	18.28
Location 2	

Table 10: In-situ DCP Result at Location 2

DCP gives much higher CBR values (18–72%) because it measures dry, in-situ strength, not soaked condition.

According to ASTM D1883, the representative CBR is taken at 2.5 mm penetration, unless the 5.0 mm value is higher, in which case the test is repeated, or the 5.0 mm value is used.

3.6.4 Determine Representative CBR Value

From Lab Result,

Table 11: Summary CBR Lab Result

Penetration	CBR %
2.5 mm	2.62
5.0 mm	3.14

Since the CBR value at 5.0 mm penetration was higher than at 2.5 mm, the test was repeated to verify the results. After confirming consistency, the CBR value at 2.5 mm penetration was adopted as the standard value, in accordance with standard practice for conservative pavement design.

Representative (Lab) CBR=2.62%

From Field Result (DCP),

Location – 01

Average Mean = $\frac{72.71+25.89+54.73+20.06+30.09}{5} = 40.7\%$

Representative CBR (Location 1) = 40.7%

Location – 02

Average Mean =$\frac{64.66+22.54+64.66+29.13+18.28}{5} = 39.85\%$

Representative CBR (Location 2) = 39.85%

Comparison Lab vs DCP

Table 12: Comparison Table Lab vs DCP

Test Method	Representative CBR (%)	Condition
Lab CBR (soaked)	2.62%	Saturated
DCP Location 1	40.7%	Dry in-situ
DCP Location 2	39.9%	Dry in-situ

The difference between Lab CBR and DCP results is mainly due to testing conditions. The Lab CBR test is usually conducted on soaked samples, representing worst-case conditions such as high moisture or rainy seasons, which reduces soil strength. In contrast, the DCP test is carried out in the field under existing (often drier) conditions, resulting in higher strength values. Therefore, Lab CBR values are generally lower and are preferred for conservative pavement design.

So, for Pavement design is governed by the saturated (soaked) condition to ensure performance during wet seasons. Therefore, the representative CBR of 2.62% is adopted.

3.6.6 Subgrade classification

Subgrade classification based on Lab CBR (2.62%)

Table 13: Subgrade Classification Table

CBR Range (%)	Classification	Typical Use
0 – 3	Very Poor	Subgrade
3 – 7	Poor to Fair	Subgrade
7 – 20	Fair	Subgrade or Sub-base
20 – 50	Good	Subgrade, Base, or Sub-base
> 50	Excellent	Subgrade, Base, or Sub-base

Source: Bowles, J.E. (1992). Engineering Properties of Soils and Their Measurement, 4th Edition. McGraw-Hill.

Based on the adopted representative CBR value of 2.62%, the subgrade is classified as very poor, since it falls below 3%. Such low strength indicates that the soil has limited load-bearing capacity and is highly susceptible to deformation under traffic loads. Therefore, appropriate improvement measures such as subgrade stabilization or the provision of a capping layer are required before pavement construction to ensure adequate performance and durability of the road structure.

<u>Location 01 :-</u>

$$\log_{10} CBR = 2.628 - 1.273 \log_{10}(DCP)$$

Section 01 :- $\log_{10} CBR = 2.628 - 1.273 \log_{10}(4)$

$= 1.8616$

$= \underline{\underline{72.71\%}}$

Section 02 :- $\log_{10} CBR = 2.628 - 1.273 \log_{10}(9)$

$= 1.4132$

$= \underline{\underline{25.89\%}}$

Section 03 :- $\log_{10} CBR = 2.628 - 1.273 \log_{10}(5)$

$= 1.7382$

$= \underline{\underline{54.73\%}}$

Section 04 :- $\log_{10} CBR = 2.628 - 1.273 \log(11)$

$= 1.3023$

$= \underline{\underline{20.06\%}}$

Section 05 :- $\log_{10} CBR = 2.628 - 1.273 \log_{10}(8)$

$= 1.4784$

$= \underline{\underline{30.09\%}}$

Figure 13: Calculations of Location 01

$$\log_{10} CBR = 2.503 - 1.15 \log_{10}(DCP)$$

Section 01 :- $\log_{10} CBR = 2.503 - 1.15 \log_{10}(4)$
$= 1.8106$
$= \underline{\underline{64.66\%}}$

Section 02 :- $\log_{10} CBR = 2.503 - 1.15 \log_{10}(10)$
$= 1.353$
$= \underline{\underline{22.54\%}}$

Section 03 :- $\log_{10} (CBR) = 2.503 - 1.15 \log_{10}(4)$
$= 1.8106$
$= \underline{\underline{64.66\%}}$

Section 04 :- $\log_{10} (CBR) = 2.503 - 1.15 \log_{10}(8)$
$= 1.4644$
$= \underline{\underline{29.13\%}}$

Section 05 :- $\log_{10} (CBR) = 2.503 - 1.15 \log_{10}(12)$
$= 1.2619$
$= \underline{\underline{18.28\%}}$

Figure 14: Calculation of Location 02

Penetration (mm)	P_t (kg)	P_s (kg)	$\frac{P_t \times C_f}{P_s} \times 100\%$	CBR %
0.5	14.27	822	$\frac{14.27 \times 1}{822} \times 100\%$	1.74%
1.0	19.06	959	$\frac{19.06 \times \overset{1}{\cancel{959}}}{959} \times 100\%$	1.99%
1.5	23.85	1096	$\frac{23.85 \times 1}{1096} \times 100\%$	2.18%
2.0	28.64	1233	$\frac{28.64 \times 1}{1233} \times 100\%$	2.32%
2.5	35.88	1370	$\frac{35.88 \times 1}{1370} \times 100\%$	2.62%
3.0	40.67	1507	$\frac{40.67 \times 1}{1507} \times 100\%$	2.70%
4.0	50.25	1781	$\frac{50.25 \times 1}{1781} \times 100\%$	2.82%
5.0	64.53	2055	$\frac{64.53 \times 1}{2055} \times 100\%$	3.14%
7.5	90.93	2630	$\frac{90.93 \times 1}{2630} \times 100\%$	3.46%
10.0	117.23	3180	$\frac{117.23 \times 1}{3180} \times 100\%$	3.69%
12.5	150.76	3600	$\frac{150.76 \times 1}{3600} \times 100\%$	4.19%

Figure 15: Calculation of Laboratory results

3.7 Pavement Design

3.7.1 Design of Traffic Loading (ESAL)

Calculate Annual Average Daily Traffic (AADT)

Peak hour volume (PHV) = 856vph

Average daily traffic (ADT) = PHV*10 = 856*10 = 8560vph

We assumed; ADT ≈ AADT. Therefore AADT = 8560vph

Calculate the percentage of axel categories

Passenger Car	- (724/856) *100 = 84.6%
Two axel single-units (4 wheel)	- (119/856) *100 = 13.9%
Two axel double-units (6 wheel)	- (13/856) *100 = 1.5%
Tandem axel (containers)	- Nil

Data for traffic loading:

- AADT in both directions – 8560vph
- The percentage of Axle categories
    - Passenger Car - 85%
    - Two axel single-units - 14%
    - Two axel double-units - 1%
    - Tandem axel – Nil
- The percentage of traffic on the design lane - 100% (f_d=1)
- Growth rate - 4% (r=0.04)
- Design life - 15 years (n=15)

- Road equivalency factors (F_{ei}) (from Table 19.5 a)
  - Passenger Car - 0.0002
  - Two axel single-units (4 wheel) - 0.002
  - Two axel double-units (6 wheel) - 0.10
  - Tandem axel – 0.088

Growth Rate (G)

$$G_{r,n} = \frac{[\ (1+r)^n - 1]}{r}$$

$$G_{0.04,15} = \frac{[\ (1+0.04)^{15} - 1]}{0.04} = 20.02$$

Calculating the AADT values ($\mathbf{AADT_i}$)

AADT for passenger cars. = 8560vph * 84.6% = 7242vph

AADT for Two axel single units. = 8560vph * 13.9% = 1190vph

Two axel double-units. = 8560vph * 1.5% = 129vph

Traffic Loading (ESAL/ $\boldsymbol{W_{18}}$)

$$ESAL = f_d \times G_{r,n} \times AADT_i \times 365 \times N_i \times F_{ei}$$

Passenger Car; $ESAL_1 = 1 \times 20.02 \times (7242) \times 365 \times 2 \times 0.0002$

$= 21167.79$

Two axel single units (4 wheel); $ESAL_2 = 1 \times 20.02 \times (1190) \times 365 \times 2 \times 0.002$

$= 34782.75$

Two axel double-units (6 wheel); $ESAL_3 = 1 \times 20.02 \times (129) \times 365 \times 2 \times 0.10$

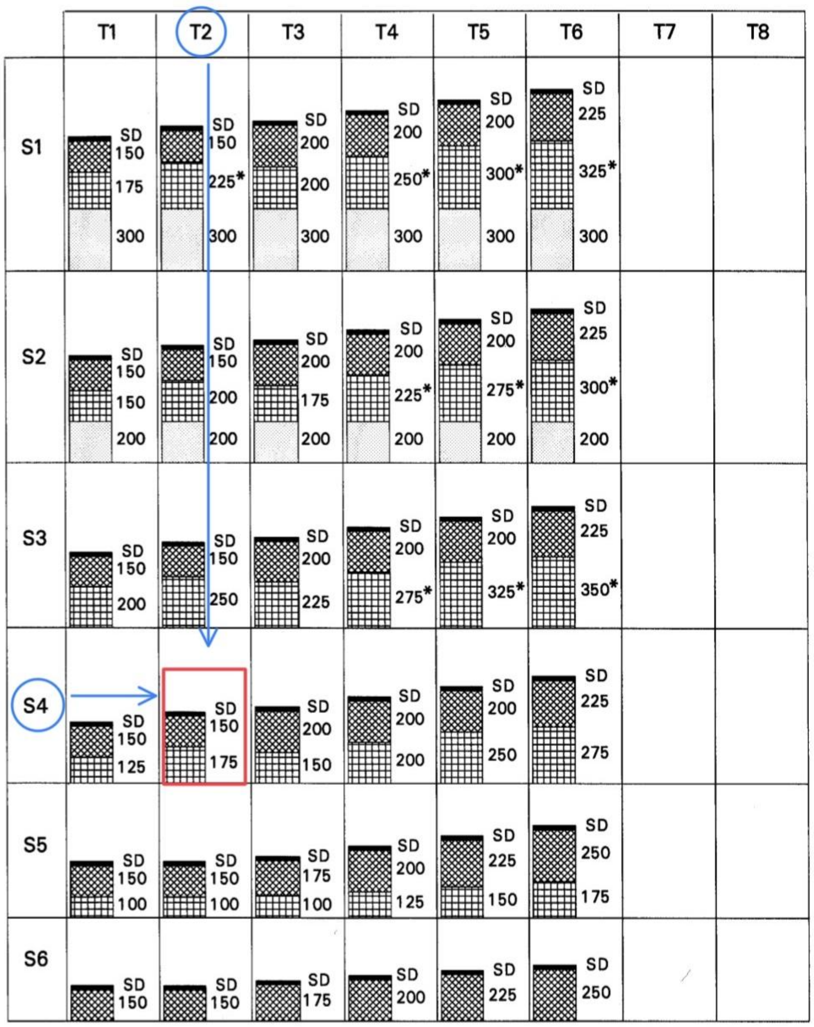
$= 188528.34$

$$\text{Total } ESAL = ESAL_1 + ESAL_2 + ESAL_3 = 21167.79 + 34782.75 + 188528.34$$

$$= 244478.88 = 0.25 \times 10^6 \approx 0.3 \times 10^6$$

- Based on the calculated total ESAL value, the traffic class can be classified as **T2**.
- When the CBR value of the subgrade is 10%, the subgrade strength class is classified as **S4**.
- Based on Chart 01, the selected pavement cross section structure corresponds to the boxed option

CHART 1 GRANULAR ROADBASE / SURFACE DRESSING



Note: 1 * Up to 100mm of sub-base may be substituted with selected fill provided the sub-base is not reduced to less than the roadbase thickness or 200mm whichever is the greater. The substitution ratio of sub-base to selected fill is 25mm : 32mm.

2 A cement or lime-stabilised sub-base may also be used.

Figure 16: Cross-sections of granular pavement structure

3.7.1 Design of Pavement

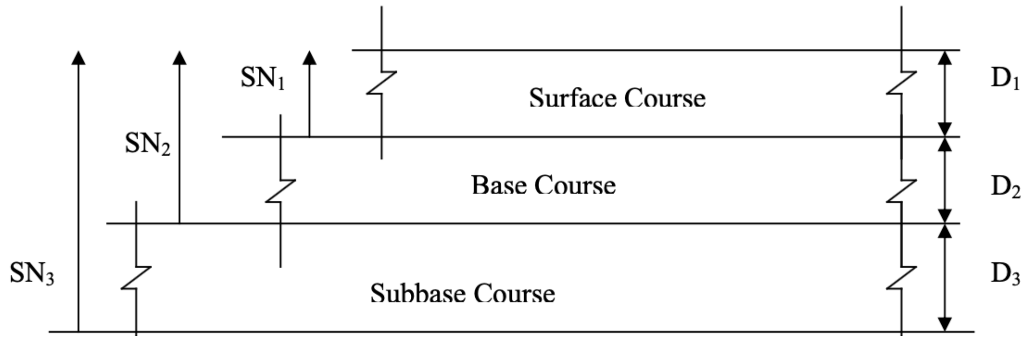


Figure 17: Structure of pavement layers

In our design we only consider the surface course (D_1) and base course (D_2). Using the AASHTO method, we designed the pavement layers. For our estimations, we obtained data from the *AASHTO Guide for Design of Pavement Structures (1993)*.

Given data:

- Traffic loading (W_{18}) – $0.3 \times 10^6 - 0.3$ millions
- Elastic modulus of the asphalt (E_{AC}) $- 400{,}000 lb/in^2$
- Growth rate - 4% (r=0.04)
- CBR of design subgrade (given) - 10%
- Design life of pavement – 15 years (n=15)
- Structural coefficient of the granular base layer (a_1) – 0.14
- All m_i values – 1
- Percentage of the traffic design lane – 40%
- Reliability level (R)– 90%
- Standard deviation – 0.4
- Serviceability loss (ΔPSI) – 1.5
- M_R (lb/in^2) – 1500CBR (For fine-gain soils with soaked CBR of 10 or less)
- M_R (lb/in^2) – 1000+555*R value (For R$\leq$ 20)

Determination of Structural Number SN_1:

- Calculating the Resilient Modulus (M_R) for base course layer

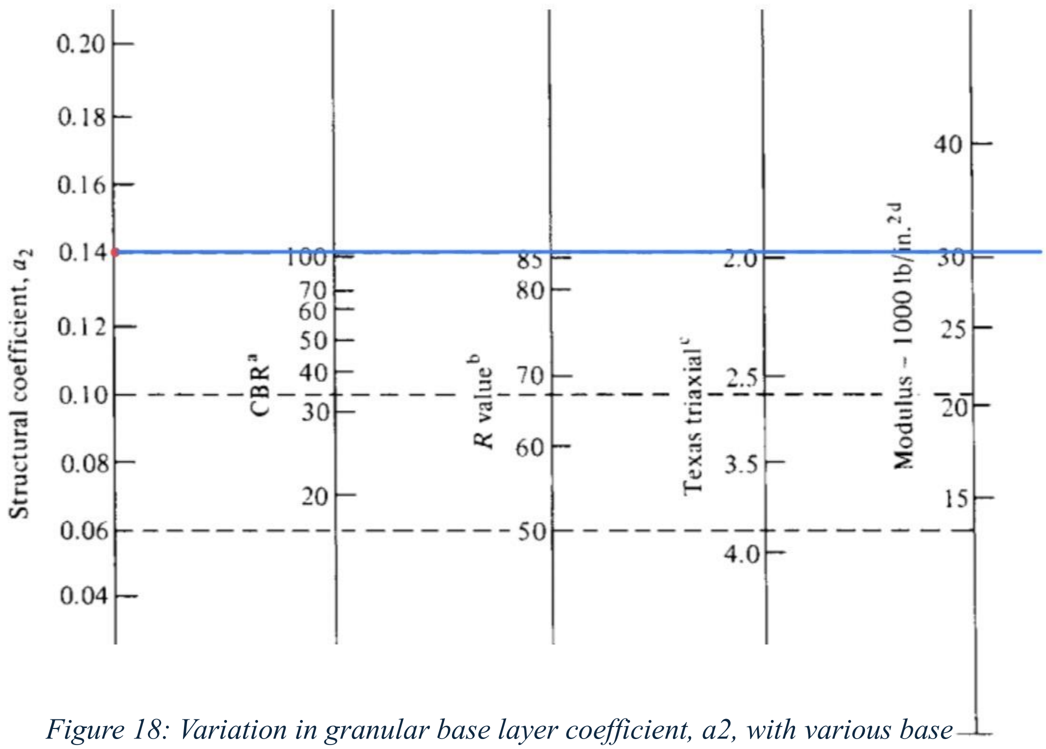


Figure 18: Variation in granular base layer coefficient, a2, with various base strength parameters

a_2 is given as 0.14 by using this value and drawing horizontal line along 0.14, we got

CBR values of base course material = 100

$$M_R = 31000psi = 31ksi$$

- $R = 90\%,\ S_o = 0.40,\ W_{18} = 0.3$ millions, $M_R = 31ksi,\ \Delta PSI = 1.5$

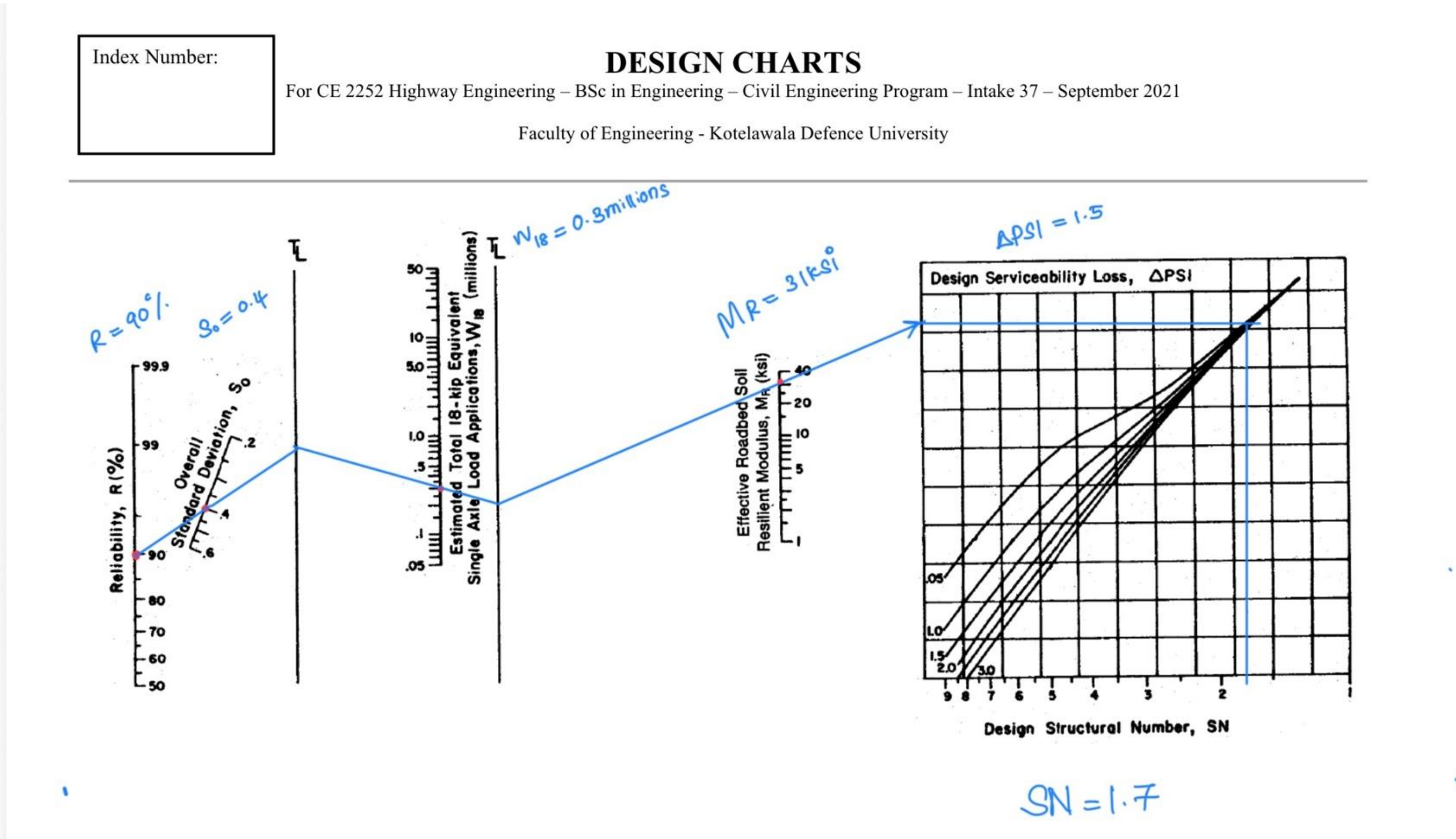


Figure 19: Design chart for SN1

By plotting the respective values on the nomograph, the structural number SN_1 was determined to be **1.7**.

Determination of Structural Number $\mathbf{SN_2}$:

- When CBR values of subgrade material is 10, we used following equation for find Resilient Modulus (M_R)

$$M_R of\ subgrade = 1500 \times 10 = 15000psi = 15ksi$$

- $R = 90\%,\ S_o = 0.40,\ W_{18} = 0.3millions,\ M_R = 15ksi,\ \Delta PSI = 1.5$

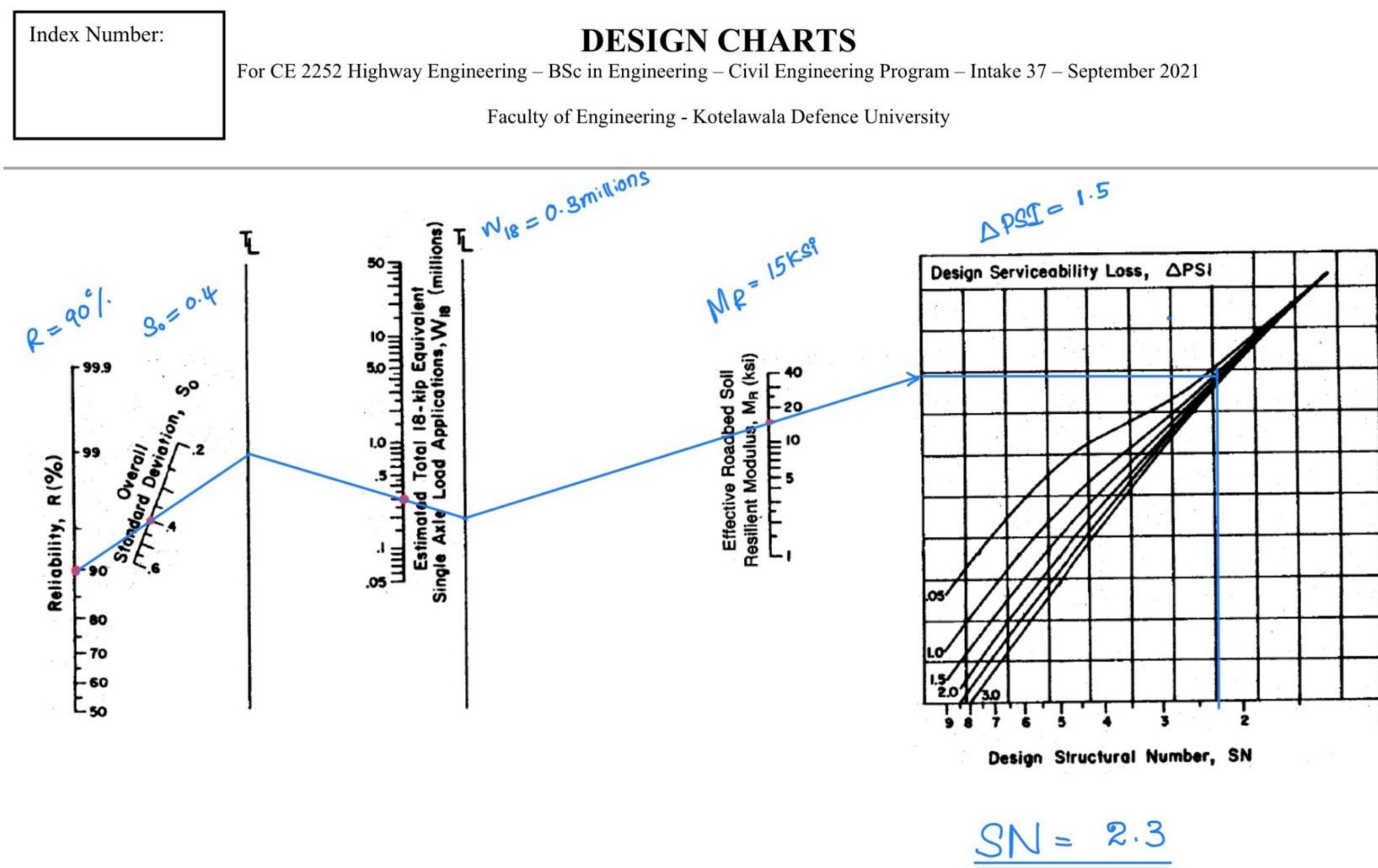


Figure 20: Design chart for SN2

By plotting the respective values on the nomograph, the structural number SN$_1$ was determined to be **2.3**.

Determination of surface layer thickness D_1;

- $SN_1 = 1.7$, $a_1 = 0.42$

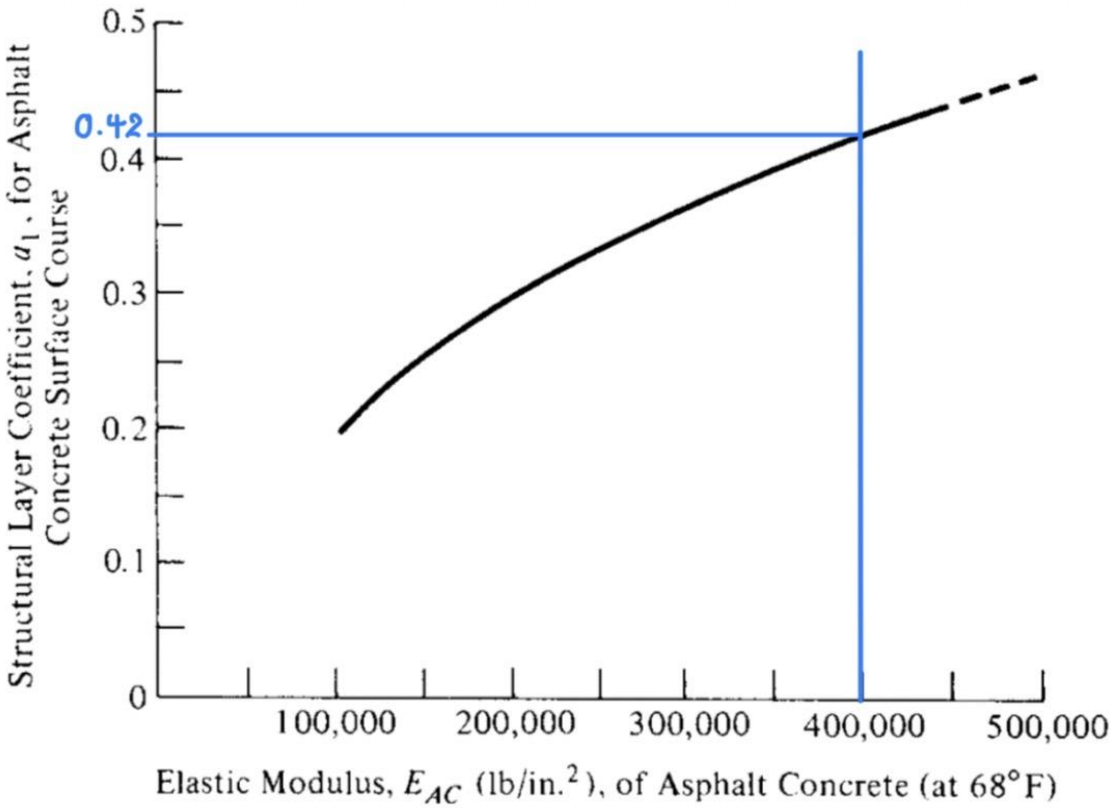


Figure 23: Chart for estimating structural layer coefficient layer a1

$$SN_1 = a_1 \times D_1$$

$$1.7 = 0.42 \times D_1$$

$$D_1 = 4.05\ in$$

$$\boldsymbol{D_1^* \approx 4\ in}$$

$$SN_1^* = 0.42 \times 4$$

$$\boldsymbol{SN_1^* = 1.68}$$

Determination of base course layer thickness D_1;

- $SN_2 = 2.3,\ SN_1^* = 1.7\ ,\ a_2 = 0.14\ (given),\ m_2 = 1(given)$

$$SN_2 = SN_1^* + a_2 \times m_2 \times D_2$$

$$2.3 = 1.68 + 0.14 \times 1 \times D_2$$

$$\boldsymbol{D_2} = \frac{2.3 - 1.68}{0.14} = 4.43in$$

$$\boldsymbol{D_2^* \approx 5\ in\ ;}$$

$$SN_2^* = SN_1^* + a_2 \times m_2 \times D_2^*$$

$$SN_2^* = 1.68 + 0.14 \times 1 \times 5$$

$$\boldsymbol{SN_2^* = 2.38}$$

Table 19.11 AASHTO-Recommended Minimum Thicknesses of Highway Layers

Traffic, ESALs	*Minimum Thickness (in.)*	
	Asphalt Concrete	*Aggregate Base*
Less than 50,000	1.0 (or surface treatment)	4
50,001–150,000	2.0	4
150,001–500,000	2.5	4
500,001–2,000,000	3.0	6
2,000,001–7,000,000	3.5	6
Greater than 7,000,000	4.0	6

SOURCE: Based on *AASHTO Guide for Design of Pavement Structures*, 1993, AASHTO, Washington, D.C.

Figure 24: Minimum thickness

Based on the recommendations of the minimum thickness table, Minimum thickness is 4in and we obtained value **satisfies the required criteria.** The pavement will therefore consist of 4 in. surface course, and 5 in. base course.

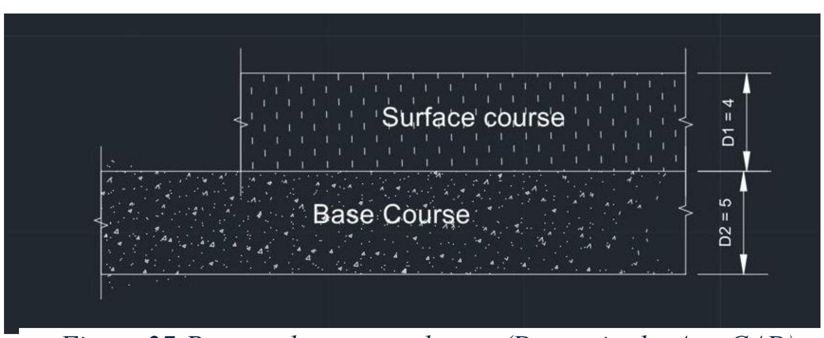


Figure 27:Proposed pavement layers (Drawn in the AutoCAD)

3.8 Geometric Design Improvements

3.8.1 Introduction

Geometric design of a road involves the arrangement of its physical features, such as lane width, shoulder width, alignment, and sight distance, to ensure safe and efficient traffic flow. Evaluating the existing road geometry helps to identify deficiencies that may affect safety and performance. Therefore, appropriate improvements are proposed based on standard design guidelines to accommodate future traffic demand and enhance the overall efficiency and safety of the road.

3.8.2 Lane Width Improvement

Existing Lane width = 3.7 m

Carriage width = 7.4 m

Problem

- Too narrow
- Difficult for buses/trucks

Proposed

3.75 m (recommended for two-lane roads)

The existing lane width is 3.75m, which is below standard. According to design guidelines, a lane width of 3.5 m is recommended. Therefore, widening is proposed to improve safety and traffic flow.

3.8.3 Shoulder Width Improvement

Existing Shoulder Width 0.5m

Problem

- No space for emergencies
- Edge damage likely

Proposed

- Minimum: 1.8-3.0 m

The existing shoulder width is inadequate. A minimum of 1.8 m is proposed to enhance safety and provide space for emergency stops.

3.8.4 Sight Distance Improvement

The available sight distance along the road is restricted in several locations due to obstructions such as roadside vegetation, buildings, and sharp curves. Adequate stopping sight distance is a critical requirement for safe driving. Therefore, it is proposed to remove or relocate these obstructions and modify the alignment where necessary to achieve the required sight distance standards.

3.8.5 Cross Section Improvement

Existing

Narrow road

Poor drainage

Problems

- Water logging
- Unsafe edges

Proposed

Include:

- Lane + shoulder
- Camber (2–3%)
- Side drains

The existing cross section lacks proper camber and drainage facilities, which can lead to water accumulation on the pavement surface and eventual deterioration. A standard cross section with adequate camber and drainage is proposed to improve surface runoff and durability. These improvements will enhance the durability of the pavement and ensure safer driving conditions during adverse weather.

3.8.6 Horizontal and Vertical Alignment

Horizontal Alignment (Curves)

Existing: a small radius curve

Problems

- Poor visibility
- Accident risk

Proposed

- Increase curve radius
- Add superelevation
- Provide extra widening

The existing horizontal curve has a small radius, resulting in poor visibility. Increasing the radius and providing superelevation will improve safety and driving comfort.

Vertical Alignment

The vertical alignment of the selected road segment does not include properly designed vertical curves, and changes in gradient occur abruptly along the road. These sudden changes can reduce driving comfort and limit sight distance, especially at crest and sag locations, creating potential safety issues for drivers. To improve this condition, it is recommended to introduce appropriate vertical curves at points where the gradient changes. This will provide a smooth transition between slopes, enhance visibility, and improve overall safety and driving comfort in accordance with standard highway design practices.

3.8.7 Summary of Geometric Improvements

Table 14: Summary table of geometric improvement

Element	Existing Condition	Proposed Improvement
1.Lane width	Narrow	3.5 m standard lanes
2.Shoulder Width	Inadequate	1.8 m shoulders
3.Cross fall	Poor drainage	2-3% camber
4.Sight Distance	Obstructed	Clear obstructions & improve alignments
5.Drainage	Limited	Improved side drains

In conclusion, proper geometric design of the road will enhance safety, visibility, and overall driving comfort.

3.9 Drawings

- Project site location: Gangarama Road
- Access roads: Borupana Road
- Surrounding land use: residential and vegetation

Figure 28: Existing Road cross section

The existing cross section represents the current condition of the road before any improvements. It shows the present width, surface condition, and side features such as shoulders and drainage. This section helps in identifying deficiencies like inadequate width, poor drainage, or uneven surfaces, which justify the need for rehabilitation or upgrading.

Data Points:

- Lane width: 3.7m
- Existing surface type: asphalt
- Shoulder width: 0.5m
- Drainage features: slopes

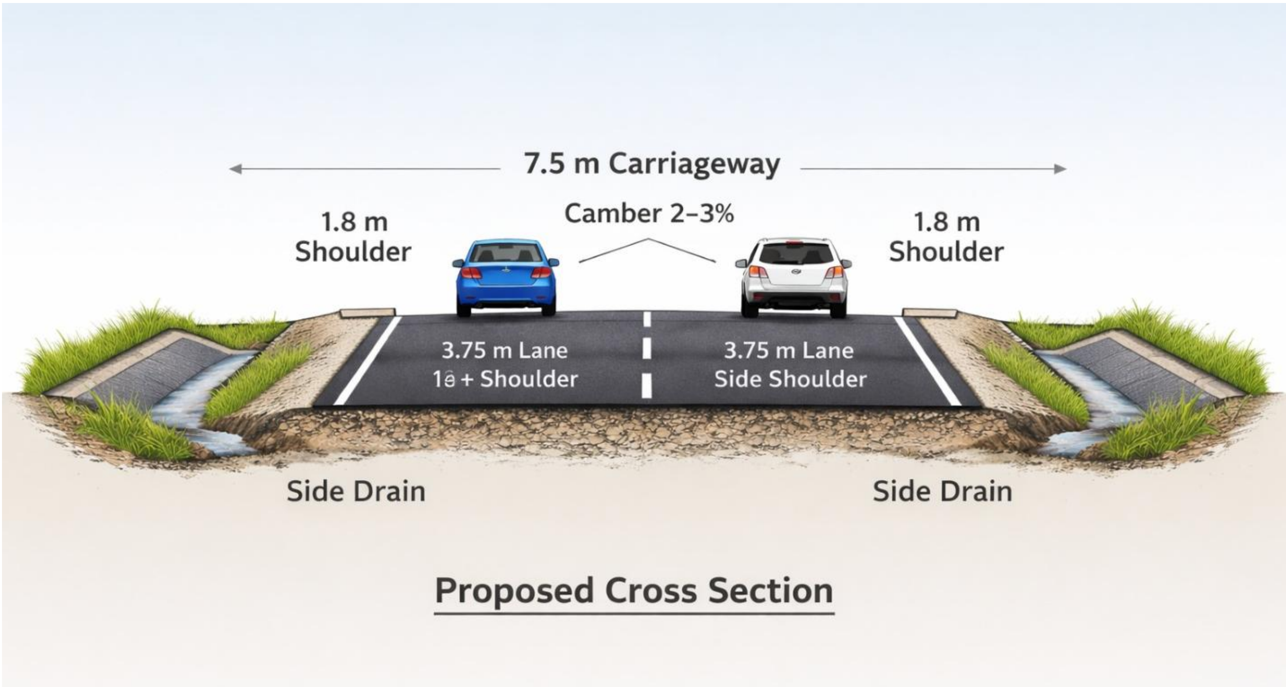


Figure 29: Proposed cross section

The proposed cross section presents the improved design of the road after modifications. It includes enhanced dimensions, better surface structure, and improved drainage systems. The design aims to meet required standards for safety, durability, and traffic efficiency while addressing the shortcomings identified in the existing condition.

Data Points:

- Proposed carriageway width: 7.5m
- Shoulder width: 1.8m
- Camber: 2-3%
- Drainage improvements: Improved side drains

4. REFERENCES

[1] American Association of State Highway And Transportation Officials (1993). *AASHTO guide for design of pavement structures*. Washington (D.C.): AASHTO

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